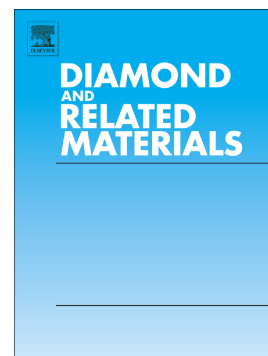


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Improved photocatalytic H₂ evolution performance of mesoporous graphitic carbon nitride with cyano-group

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Abstract

In this paper, two modification strategies were introduced in g-C₃N₄ system to improve its photocatalytic activity. In detail, mesoporous g-C₃N₄ with cyano-group was successfully synthesized. Its chemical structure and states, specific area, optical absorption, photo-electrochemical properties and photocatalytic ability for hydrogen evolution were characterized. It was indicated that mesoporous g-C₃N₄ with cyano-group photocatalyst exhibited better charge separation rate, and the higher photocurrent intensity. For hydrogen evolution, it achieved the higher hydrogen evolution rate of 4531 $\mu\text{mol}\cdot\text{h}^{-1}\text{g}^{-1}$, which was 3 folds that of mesoporous g-C₃N₄ and 13 folds that of bulk g-C₃N₄, and kept excellent stability. Obviously, current work shed light for developing high efficient photocatalysts by combining multiple modification.

Keywords: Photocatalysis; Mesoporous g-C₃N₄; Hydrogen evolution; Multiple modification

1. Introduction

Semiconductor photocatalytic technology can directly use sunlight to split water to produce H_2 and degrade various kinds of pollutants, which is one of the most ideal ways to supply green energy [1-7]. It is essential to develop robust photocatalysts with high stability, low cost and good efficiency. As a new generation of nonmetallic semiconductor photocatalyst, g- C_3N_4 was reported to catalyze hydrogen and oxygen production from water, degrade pollutants, CO_2 reduction, etc, and had some advantages, including good chemical stability, wide material source and visible light response [8-15]. Usually, g- C_3N_4 photocatalyst could be simply produced *via* the calcination of precursors with nitrogen (melamine, dicyandiamide, cyanamide, etc.). g- C_3N_4 prepared from this method existed some drawbacks (small specific area, fast charge carrier recombination rate), which resulted that its photocatalytic performance is not high enough. Apparently, g- C_3N_4 materials with high photocatalytic activity could be successfully prepared by overcoming these shortcomings.

At present, to solve the shortcomings of g- C_3N_4 material, a great many of scholars have developed a series of effective strategies to improve its photocatalytic properties, such as building heterojunction with other materials [16-20], changing morphology [21-23], doping other elements [24, 25], introducing porous structure [26-28], improving crystallinity [29-31], establishing defect engineering [32-35], etc. Nevertheless, one of above methods cannot simultaneously modify all of shortcomings of g- C_3N_4 , for example, in previous our group work [36], g- C_3N_4 with cyano-group was prepared, its separation rate of photo-induced charges was improved,

which boosted its H_2 production ability, but its surface area was only $7.6 \text{ m}^2/\text{g}$, it existed a large improved space. Hence, the consociation of various methods is necessary. Previous reports also proved this result. Mesoporous structure and elements doping (S, O, C) were simultaneously employed to modify $g\text{-C}_3\text{N}_4$, their synergistic effect endowed the prepared photocatalysts with high enhanced photocatalytic ability [37-40]. Obviously, the combination of various methods could further increase the catalytic property of $g\text{-C}_3\text{N}_4$.

In current work, mesoporous $g\text{-C}_3\text{N}_4$ with cyano-group was synthesized. Its chemical structure and states, specific area, optical absorption, photoelectrochemical properties and the photocatalytic ability for H_2 evolution were characterized. It is observed that as-prepared mesoporous $g\text{-C}_3\text{N}_4$ with cyano-group photocatalyst exhibits superior photocatalytic activity over mesoporous or bulk one. Current work provides an effective method of developing high efficient $g\text{-C}_3\text{N}_4$ by combining multiple modifications.

2. Experimental methods

2.1 The synthesis of catalysts

4 g dicyandiamide was firstly added in 20 mL deionized water. Subsequently, 2 g silicon dioxide (15 nm) was added under persistently stirring. After 2 h. Then deionized water was removed at 80°C by evaporation. The mixture were stayed in a covered crucible and heated at 550°C for 2 h to gain $\text{SiO}_2/g\text{-C}_3\text{N}_4$.

1g of $\text{SiO}_2/g\text{-C}_3\text{N}_4$ was adequately ground with various mass KCl and LiCl

molten salt (the weight ratio of KCl and LiCl was 11:9) in agate mortar for 20 min. Then, the mixture were treated at 550 °C for 2 h under nitrogen atmosphere. After cooling down, molten salt was washed by plenty of hot deionized water. Silicon dioxide was removed by etching with hydrofluoric acid solution (60 mL, 7 wt%) for 24 h. According to the added mass of molten salt (1 g, 3 g and 5 g), the samples were marked as mDCN-M-1, mDCN-M-2 and mDCN-M-3. As compared sample, the preparation of the mesoporous g-C₃N₄ (mDCN) was obtained using the same procedures except the addition of KCl and LiCl. Bulk g-C₃N₄ was produced via heating 4 g dicyandiamide at 550 °C for 120 min and marked as DCN.

2.2 Characterizations and H₂ evolution

The details were supplied in Supporting information.

3. Results and Discussion

Fig. 1A presents the XRD patterns of resulted mDCN and a series of mDCN-M samples to characterize their crystalline nature. For mDCN, the stronger peak (27.5°) assigned to the layered stacking of heptazine frameworks [41]. Another one (13.1°) attributed to the distance between the tri-s-triazine structures with a certain periodicity. For mDCN-M, the stronger peak was shifted to slightly larger angle than mDCN, this result indicated the reduced interlayer distance. Furthermore, the weak peak was around 8.1° in mDCN-M sample, which derived from the increased in-plane periodicity, which might attribute to the intercalation of K⁺ during the post-treated process in the molten salt system [42]. Fig. 1B shows the FTIR spectra of mDCN and

a series of mDCN-M. The overall spectrum of mDCN-M were similar to that of mDCN. Three regions were observed, including 810 cm^{-1} , $1100\text{-}1800\text{ cm}^{-1}$ and $3000\text{-}3500\text{ cm}^{-1}$, which respectively characterized the stretching vibration of triazine, CN heterocycles, and O-H and N-H. Moreover, these results also meant that pores introduction did not affect the chemical structure of g-C₃N₄. Interestingly, compared with mDCN, an obvious peak was observed at 2186 cm^{-1} after the materials were treated by molten salt, which was stretching vibration of cyano group [43]. Obviously, cyano group has been introduced in the g-C₃N₄ chemical structure, which attributed to the role molten salt. When g-C₃N₄ was heat treated in the molten salt, molten salt could catalyze the cleavage of g-C₃N₄, introducing the formation of cyano group [43]. Then the C, H, N element analysis for as-prepared photocatalysts was measured in Table S1. It was found that C/N ratio was lower than ideal C₃N₄ (0.75), and H also existed in materials, which meant that as-prepared sample tend to be melon structure.

(Fig. 1)

In addition, to further distinguish chemical states of various elements, XPS was also investigated. Two typical peaks were observed for C 1s spectrum (Fig. 2A). The former was the outside elements of C. The latter (288.2 eV) belonged to sp^2 -hybridized C=N. As revealed in Fig. 2B, the XPS spectrum of N 1s, which was fitted into 3 binding energies, where 398.5 eV and 399.2 eV corresponded to N in heptazine units (C-N=C) and N-(C)₃, respectively [44, 45]. 400.5 eV corresponded to amino groups. As revealed in Fig. S1 for K 2p, two binding energy were at 293.0 eV and 295.7 eV , which may be the surface bonded or layered intercalated K⁺ [46],

which agreed with the result of XRD.

(Fig. 2)

The features of as-synthesized mDCN and mDCN-M-2 were measured *via* SEM. In Fig. 3A-B, mDCN and mDCN-M-2 were exhibited massive, dense and bulk structure. To further study their microstructure, the TEM observations were carried out and listed in Fig. 3C-D. For mDCN in Fig. 3C and mDCN-M-2 in Fig. 3D, irregular lamellar structure could be seen and a great number of pores (labeled by red circles) were found. And the TEM image of bulk g-C₃N₄ was also supplied in Fig. S2, clearly, bulk g-C₃N₄ had smooth surface, by comparison, it was found that mDCN and mDCN-M-2 presented the rough surface, which was resulted from the introduced porous structure. In this system, the presence of pores increases their surface area and active sites, which helps to improve the catalytic capacity.

(Fig. 3)

In order to further prove that the prepared samples are porous materials, nitrogen adsorption-desorption curves were tested in Fig. 4A. All photocatalysts exhibited a typical type IV isotherm, showing the characteristics of porous materials. Meanwhile, the pore size distributions of resulted photocatalysts were shown in Fig. 4B, the pore diameter of all sample is focus on 11.7 nm, which was agreed with the SiO₂ templates. The specific surface areas of mDCN, mDCN-M-1, mDCN-M-2 and mDCN-M-3 were 85.6, 50.9, 48.4 and 48.1 m² g⁻¹, respectively.

(Fig. 4)

The light absorption capacity is also a crucial condition to determine its

photocatalytic activity of photocatalyst. Therefore, the absorption of mDCN-M-2 and mDCN samples was determined by DRS. As presented in Fig. 5A, it was clearly observed that the visible light absorption band edge of mDCN sample was 460 nm, compared with the mDCN, after being modified by molten salt, the absorption of mDCN-M-2 was significantly enhanced between 500 and 800 nm, and the visible absorption wavelength blue shifted to 440 nm. Using Kubelka-Munk method, band gap (E_g) of mDCN and mDCN-M-2 were also gained in Fig. 5B. The E_g of mDCN and mDCN-M-2 were respective 2.52 and 2.68 eV. This indicated that band gap the prepared mDCN-M had blue shift after mDCN was post-treated by molten salt, and showed enhanced visible light response capacity.

(Fig. 5)

Then electron paramagnetic resonance (EPR) spectra was recorded (Fig. 6). Compared with mDCN, the EPR signal intensity of mDCN-M-2 has clearly increased. This phenomenon should be attributed to the unpaired symmetric electrons exist in the modified carbon nitride.

(Fig. 6)

Fig. 7A shows the fluorescence spectra of mDCN and mDCN-M-2 samples at room temperature. As shown in the figure, it was obvious that mDCN had a high emission signal under the same excitation wavelength, indicating that more carriers were involved in the recombination process. Compared with mDCN, the mDCN-M-2 showed a significant reduced intensity of the fluorescence signal, indicating the reduced recombination probability of photogenic electrons and holes [47,

48]. Furthermore, transient photocurrent responses of mDCN and mDCN-M-2 were also measured and presented in Fig. 7B. Under the same conditions, two samples displayed typical photocurrents, the photocurrent response of mDCN-M-2 was obviously better than mDCN, which showed that the mDCN-M-2 had higher separation property of photo-induced charge. Hence, above results firmly confirmed that mDCN-M had higher separation capacity of photo-induced charge.

(Fig. 7)

The photocatalytic ability for H₂ evolution over prepared photocatalysts was shown in Fig. 8A. As light irradiated time lasted longer, all the samples produced hydrogen continuously, but exhibited different rates. By comparison, mDCN-M-2 produced the most H₂ amount under the same time. Fig. 8B exhibits hydrogen evolution rate (HER) of prepared samples, compared with DCN (346 $\mu\text{mol}\cdot\text{h}^{-1}\text{g}^{-1}$) and mDCN (1506 $\mu\text{mol}\cdot\text{h}^{-1}\text{g}^{-1}$), mDCN-M-1, mDCN-M-2 and mDCN-M-3 had better hydrogen production capacity, and HER was 1942, 4531 and 2796 $\mu\text{mol}\cdot\text{h}^{-1}\text{g}^{-1}$, respectively. mDCN-M-2 had the best hydrogen production capacity, and its HER was 3 folds that of mDCN and 13 folds that of DCN. Then the HER of obtained samples in previous reports was shown Table 1. By comparison, it was found that mDCN-M-2 presented obviously improved H₂ production ability. This reveals that mDCN-M is an excellent photocatalyst.

(Fig. 8)

(Table 1)

To investigate the continuous used ability of photocatalyst, resulted mDCN-M-2

catalyst was used to conduct multiple photocatalytic hydrogen production cycle experiments in Fig. 9. After 12 h of continuous reaction, hydrogen production capacity of mDCN-M-2 under visible light did not decrease significantly. The above results prove that mDCN-M-2 is an efficient and stable photocatalyst.

(Fig. 9)

Furthermore, the spent catalyst was further detected by XRD and FTIR. By comparison, XRD diffraction and stretching vibration peaks mDCN-M-2 samples before and after three times cycling experiments showed no significant change, manifesting that the structure of catalyst was steady in photocatalytic reaction process.

(Fig. 10)

To further understand the relation of structure and performance, electronic band structure of mDCN and mDCN-M-2 could be obtained in Fig. S4, based on VB XPS (Fig. S3) and band gap (Fig. 5) of mDCN and mDCN-M-2. Obviously, they showed the similar VB potential, by comparison, mDCN-M-2 had a more negative potential than mDCN, which revealed that mDCN-M-2 presented stronger reducing capacity. Therefore, specific electronic band structure prompted mDCN-M-2 exhibiting the improved H₂ evolution rate.

4. Conclusion

In summary, mesoporous g-C₃N₄ with cyano-group has been synthesized. The FTIR verified the introduction of cyano-group. The adsorption-desorption isotherms showed its mesoporous structure. The prepared mesoporous g-C₃N₄ with cyano-group

presented higher photocurrent density and lower fluorescence intensity, which meant the higher separation rate of photo-induced charge. For hydrogen evolution, mesoporous g-C₃N₄ with cyano-group achieved the higher HER of 4531 $\mu\text{mol}\cdot\text{h}^{-1}\text{g}^{-1}$, and kept prominent stability. Current work gives the significance of combined various modified methods for farther boosting photocatalytic property of g-C₃N₄-based materials

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Table 1 Comparison of HER over mDCN-M-2 and other g-C₃N₄ photocatalysts in previous reports

Samples	Light source	Co-catalyst loading	HER $\mu\text{mol g}^{-1}\text{h}^{-1}$	Reference
mDCN-M-2	300 W Xe Lamp ($\lambda > 400$ nm)	3 wt% Pt	4531	This work
Porous defect g-C ₃ N ₄	300 W Xe Lamp ($\lambda > 420$ nm)	3 wt% Pt	2092	[33]
g-C ₃ N ₄ with Na and cyano-group	300 W Xe Lamp ($\lambda > 400$ nm)	3 wt% Pt	985	[36]
Tubular g-C ₃ N ₄ isotype heterojunction	300 W Xe Lamp ($\lambda > 420$ nm)	3 wt% Pt	1575	[49]
Porous g-C ₃ N ₄	300 W Xe Lamp ($\lambda > 400$ nm)	3 wt% Pt	958	[50]
N-vacancy g-C ₃ N ₄	300 W Xe Lamp ($\lambda > 400$ nm)	3 wt% Pt	3147	[51]
Phosphorus-doped g-C ₃ N ₄ nanosheets	300 W Xe Lamp ($\lambda > 420$ nm)	3 wt% Pt	1211	[52]

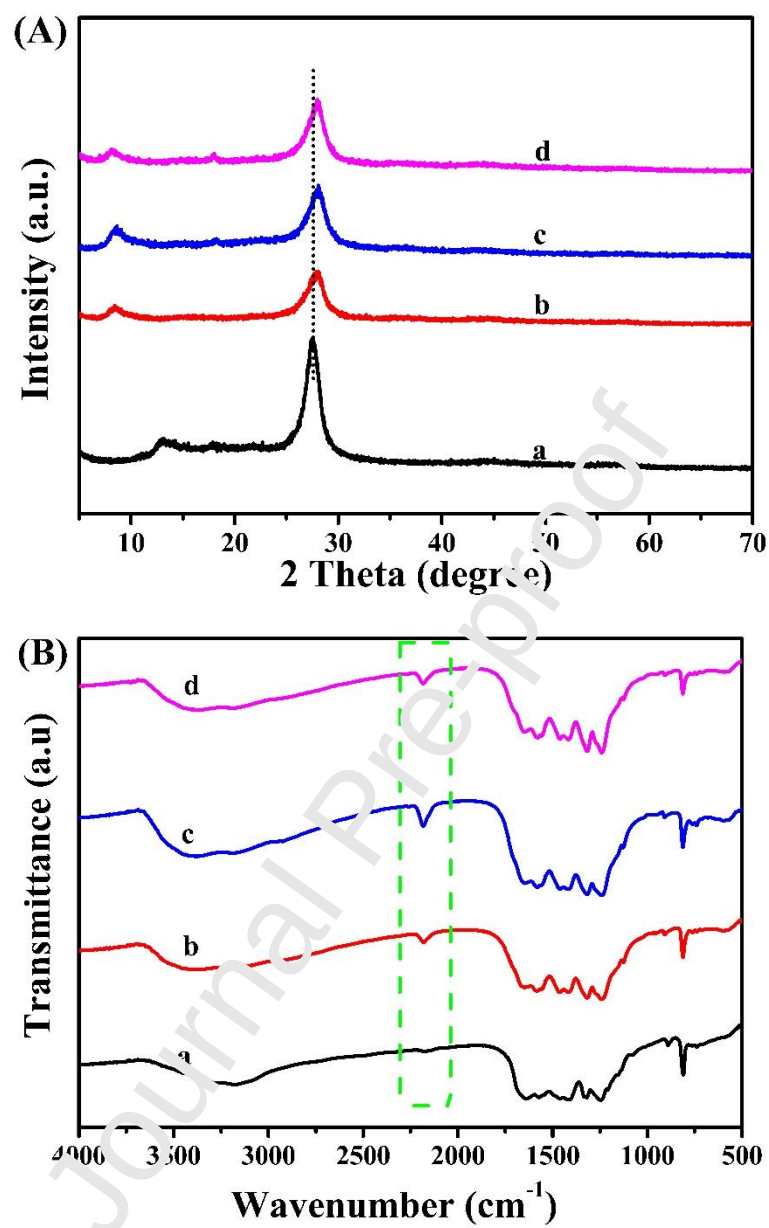


Fig. 1 (A) XRD patterns and (B) FTIR spectra of (a) mDCN, (b) mDCN-M-1, (c) mDCN-M-2 and (d) mDCN-M-3

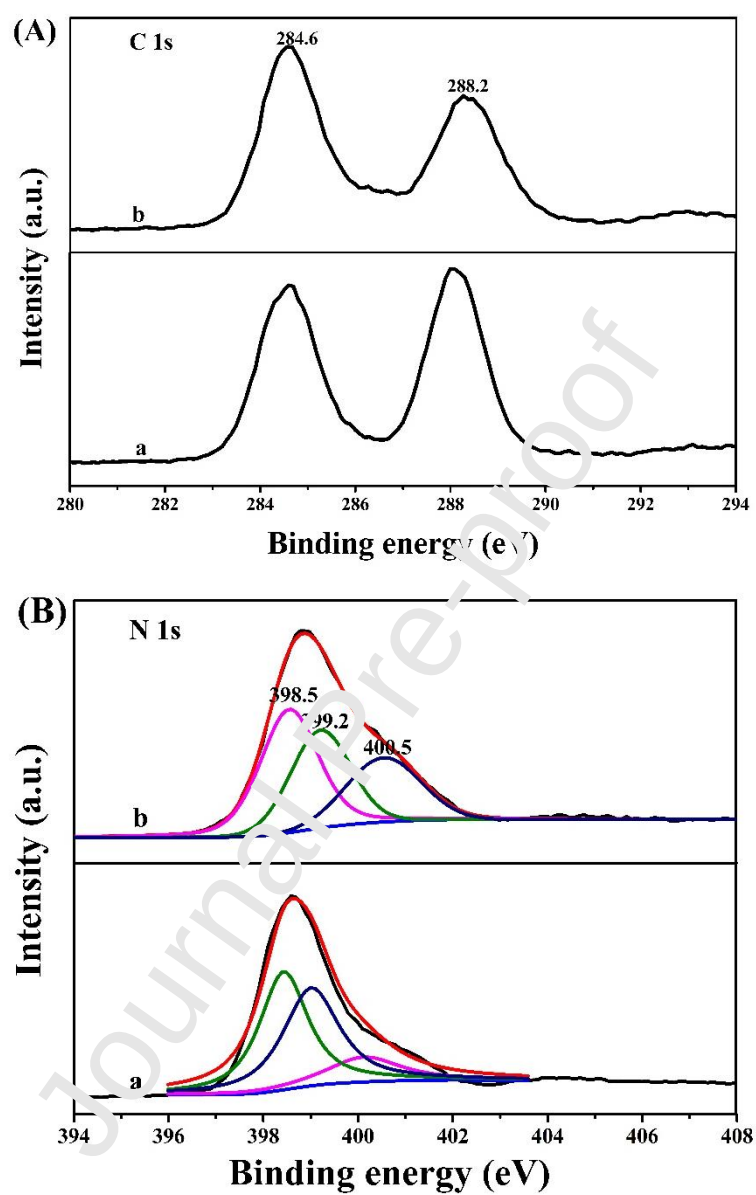


Fig. 2 XPS spectra of (a) mDCN and (b) mDCN-M-2: (A) C 1s and (B) N1s

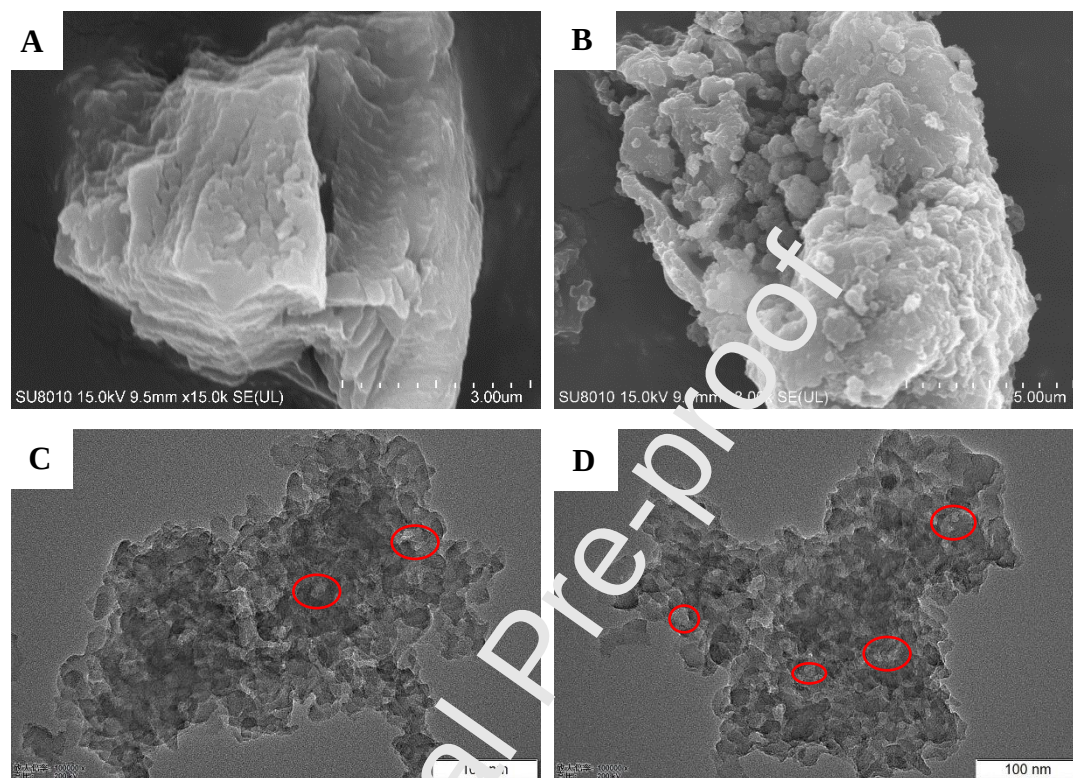


Fig. 3 The SEM images of (A) mDCN and (B) mDCN-M-2, The TEM images of (C) mDCN and (D) mDCN-M-2.

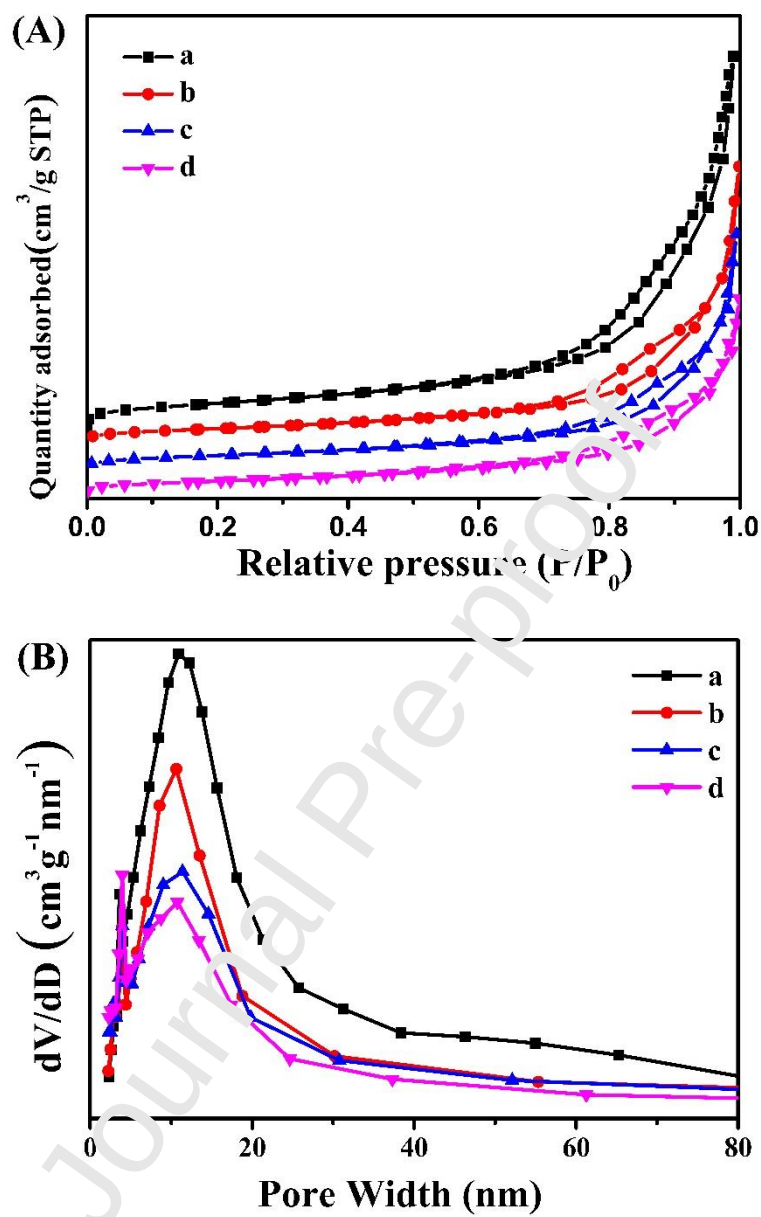


Fig. 4 (A) N_2 adsorption-desorption isotherms and (B) pore size distributions (a) mDCN, (b) mDCN-M-1, (c) mDCN-M-2 and (d) mDCN-M-3

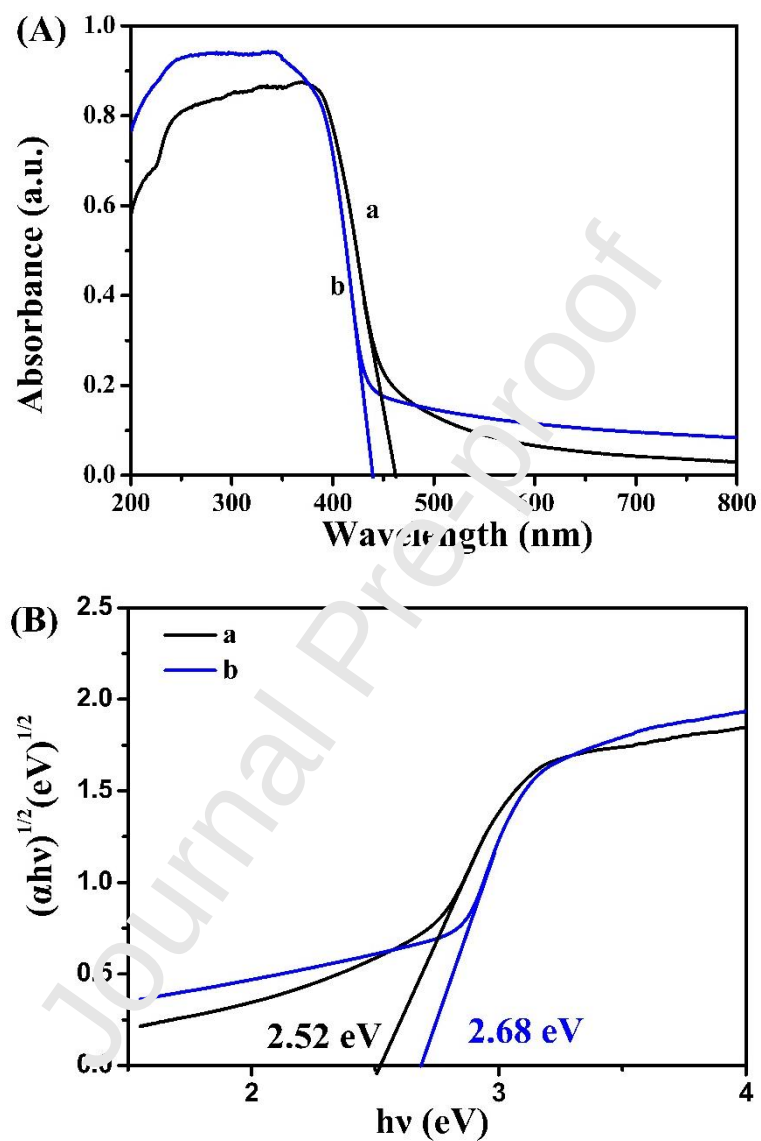


Fig. 5 (A) UV-vis diffuse reflectance spectra and (B) The curves of $(\alpha h\nu)^{1/2}$ vs. photon energy ($h\nu$) over (a) mDCN and (b) mDCN-M-2.

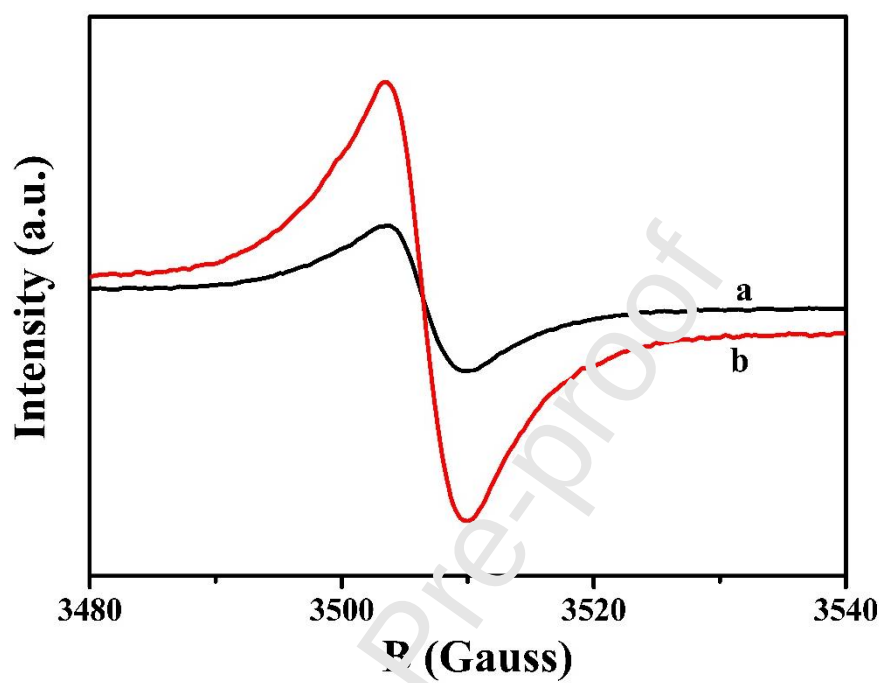


Fig. 6 The EPR spectra of (a) mDCN and (b) mDCN-M-2

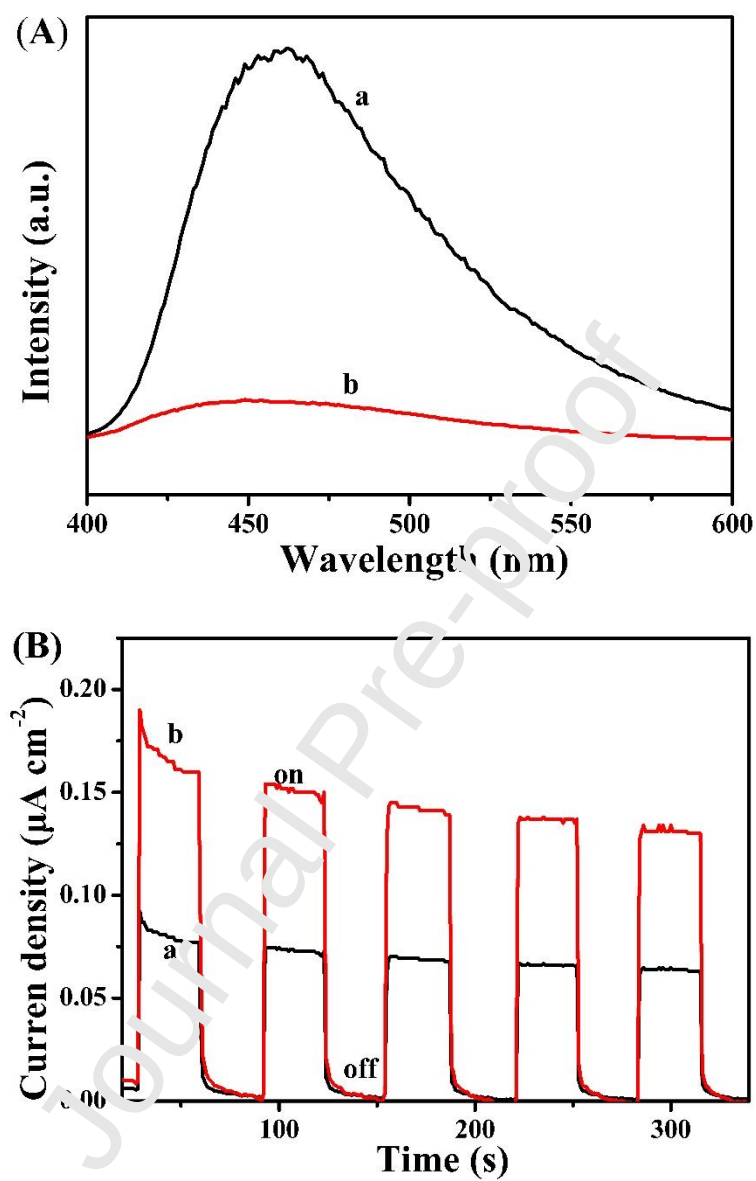


Fig. 7 (A) Photoluminescence curves and (B) Transient photocurrent curves of (a) mDCN and (b) mDCN-M-2

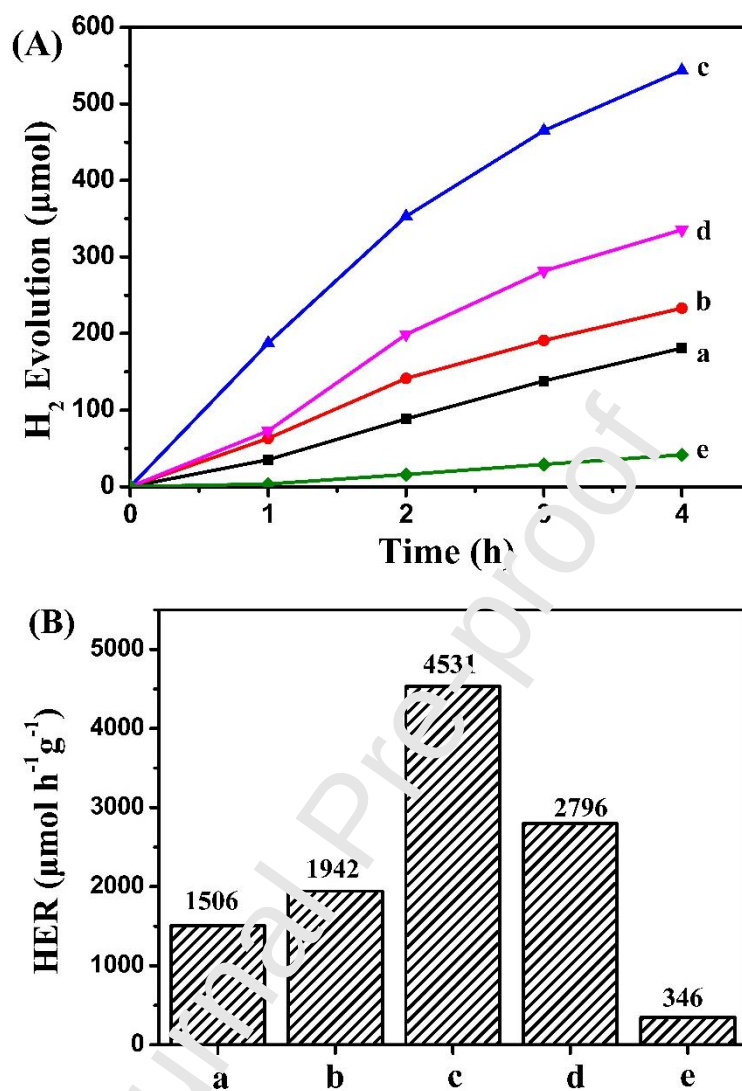


Fig. 8 (A) The curves of producing H₂ via reaction time and (B) Hydrogen evolution rates over various samples. (a) mDCN, (b) mDCN-M-1, (c) mDCN-M-2, (d) mDCN-M-3 and (e) DCN

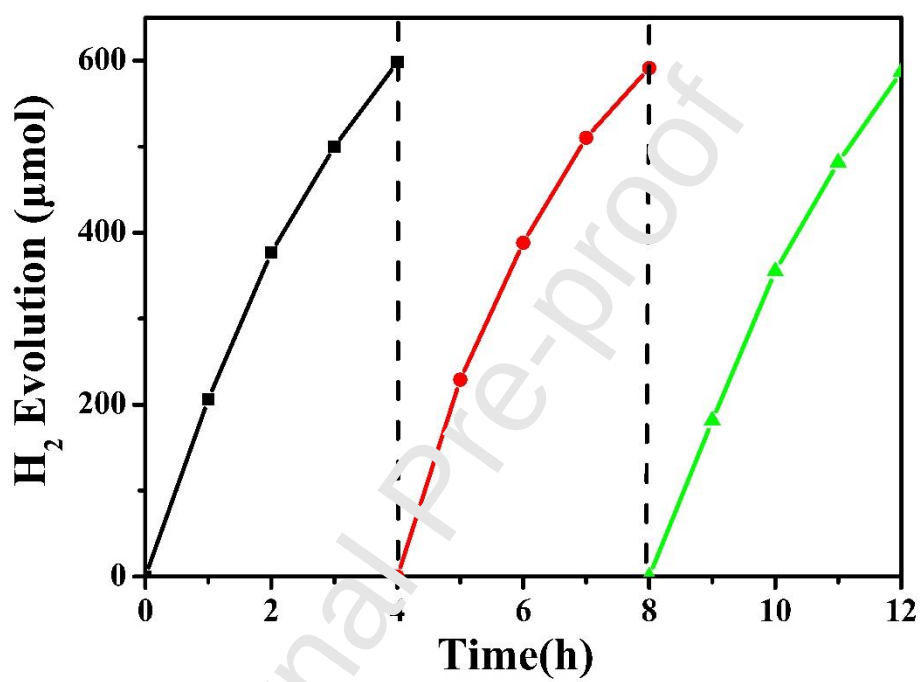


Fig. 9 Stability test in H₂ photosynthesis over mDCN-M-2

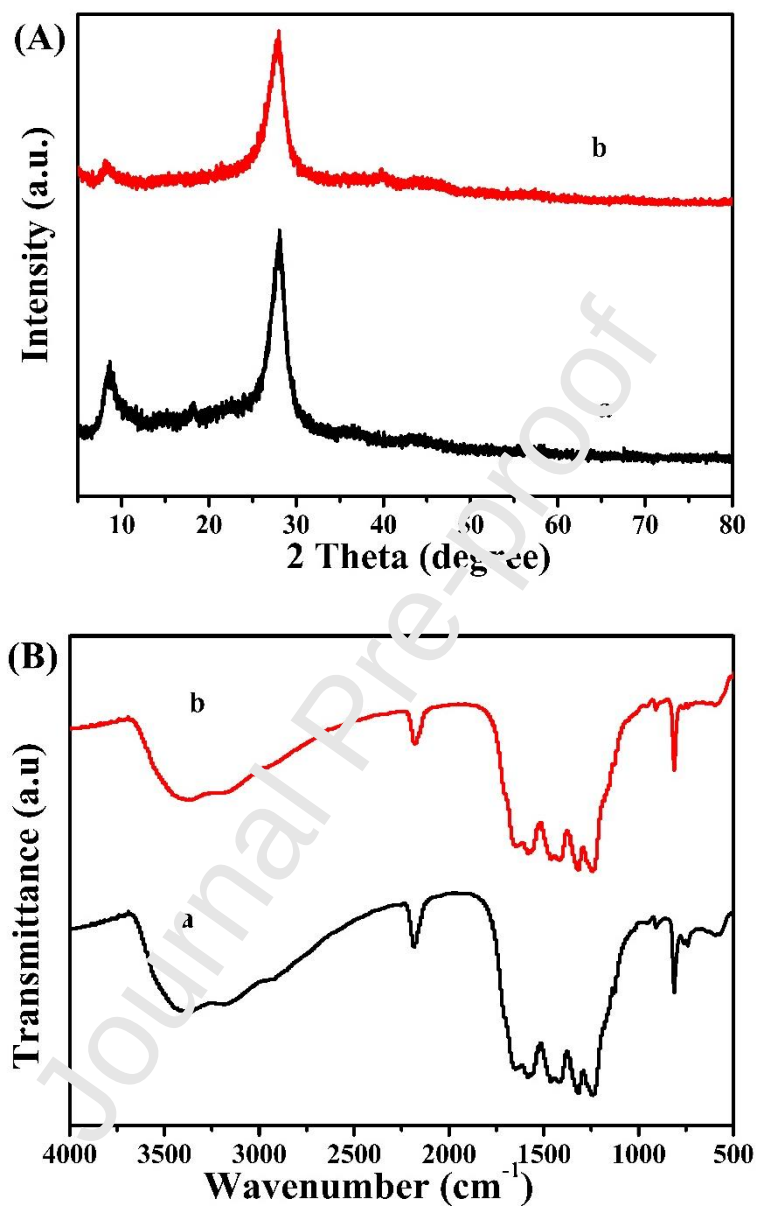


Fig. 10 (A) XRD patterns and (B) FTIR spectra of spent mDCN-M-2 (a) before and (b) after three times recycle

Shuang Li: Methodology, Software, Writing-Original draft.

Weiwei Si: Data curation, Software.

Lishuang Cui: Data curation, Software.

Lei Shi: Conceptualization, Writing-Reviewing and Editing.

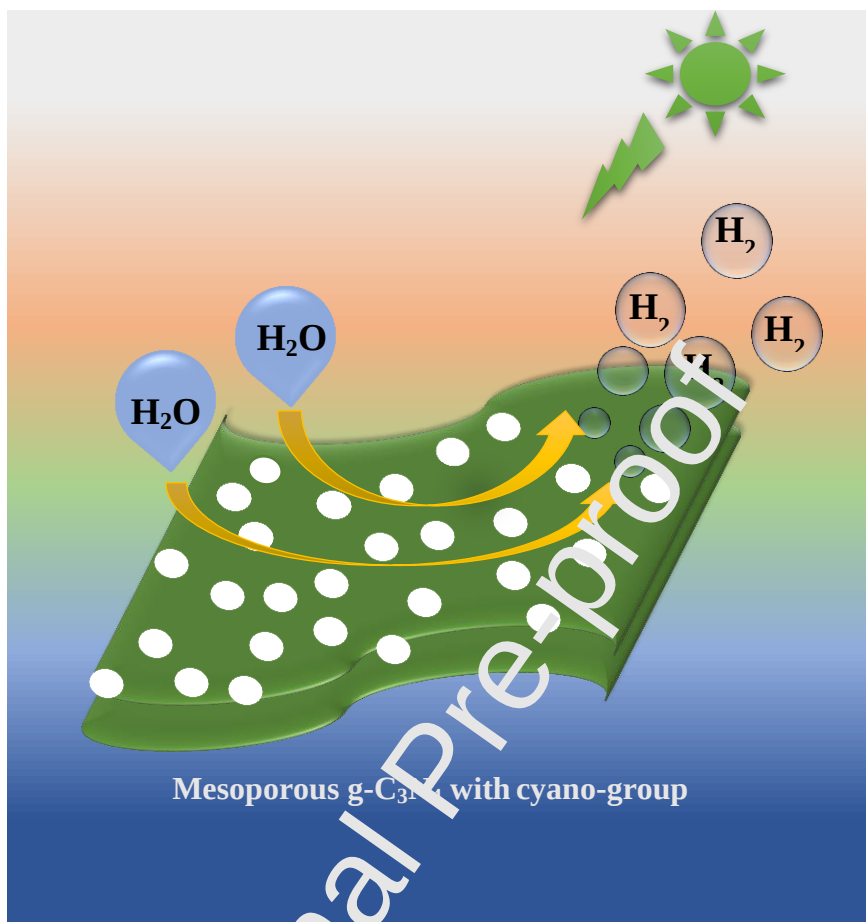
Journal Pre-proof

Conflicts of interest

There are no conflicts of interest to declare.

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Graphical abstract



Highlights

- Two modified methods are introduced in g-C₃N₄ to improve its photocatalytic ability.
- Mesoporous g-C₃N₄ with cyano-group has good separated rate of photoinduced charge.
- Mesoporous g-C₃N₄ with cyano-group exhibits large surface area.
- Mesoporous g-C₃N₄ with cyano-group exhibits excellent photocatalytic activity.
- Mesoporous g-C₃N₄ with cyano-group keeps excellent stability.